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Process mining in supply chain management: state-of-the-art, use cases and research outlook

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ABSTRACT

Recent advances in process mining technology have extended its applicability beyond traditional domains such as healthcare, finance and manufacturing, making it increasingly relevant for addressing problems in supply chain management. In our research, we explore the integration of process mining techniques within the domain of supply chain management, focusing on uncovering inefficiencies, ensuring compliance, and identifying opportunities for improvement. Therefore, we first review the technological advances in process mining relevant to supply chain management and outline six relevant approaches. We then combine the identified methodologies with expert interviews to derive and validate six specific use cases where process mining can significantly contribute to supply chain efficiency and resilience. The paper presents a detailed description of these use cases, demonstrating how process mining can provide actionable insights for a wide range of supply chain problems. We discuss the implications of our findings for practitioners, who benefit from enhanced visibility and optimisation opportunities, and researchers, who are provided with a roadmap for exploration of this promising interdisciplinary field. To the best of our knowledge, this is the first work to explore potential use cases for process mining in a supply chain context, providing a comprehensive perspective on the potential benefits and challenges.

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1. Introduction

Supply chain management (SCM), as defined by Cooper, Lambert, and Pagh (1997), is the ‘integration of business processes from end user through original suppliers that provides products, services and information that add value for customers’. Research on SCM investigates the issues and problems that occur when managing these processes in the real world (Karlsson 2016). Many well-established approaches have been used in SCM research to discover and model phenomena present in the processes within and across companies’ supply chains.

Discovering processes in the SCM context has traditionally been a manual task involving production site visits, interviews, and pen-and-paper modelling (Sunk et al. 2017). However, with the increasing availability of large data sources from enterprise resource planning (ERP), material requirements planning (MRP) systems, and sensor data, the use of advanced data analytics techniques has moved into the focus of SCM research (Hofmann 2017; Ivanov, Dolgui, and Sokolov 2019; Tiwari et al. 2024). In the operations management community, this is also discussed under the terms ‘digital footprint’ or ‘digital dust’ (Angelopoulos et al. 2023; Guha and Kumar

2018). Mature data-based techniques such as data mining and machine learning for statistical analysis, visualisation, and forecasting are supported by many existing SCM tools. However, while advanced analytics enable deep factual insights, existing techniques lack the ability to reflect interconnected events in processes along the supply chain (Van Der Aalst 2016).

Modelling and simulation have been widely applied in SCM research to imitate the behaviour of supply chains and analyse phenomena such as the bullwhip or the ripple effect (Chatfield 2013; Ivanov 2017; Wangphanich, Kara, and Kayis 2010). Such simulation models are traditionally built manually, with the user defining the model and parameters that control the likelihood of certain scenarios occurring. Although this method has been successfully used in research, it is resource-intensive and usually static, which limits its applicability in practice (Van Der Aalst 2018). Recently, the term ‘digital twin’ has been discussed in the SCM field, proposing a new solution where simulation models are based on big data stored in operational systems (Ivanov 2024; Tozanli and Saénz 2022). However, in the highly dynamic and complex SCM domain, a structured approach is required to transfer big

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data into a format, essentially a model, that forms the basis for simulation (Friederich et al. 2022).

A potential answer to the limitations that advanced data analytics and simulation face in the SCM domain could be based on a family of methods summarised under the term process mining. Process mining is a technique originating from computer science that combines data science and business process management (Van Der Aalst 2016). Essentially, the idea is to use recoded event log data to extract process models that accurately describe the true behaviour of a process. Process mining traditionally consists of four key concepts: process discovery, conformance checking, process reengineering, and operational support (Van Der Aalst 2019). In addition, there is active research at the interfaces of process mining with machine learning and simulation which are two techniques that have already proven successful in SCM (Di Francescomarino et al. 2018). This allows machine learning models to be enriched with encoded process knowledge and simulation models to be mined directly from operational data without extensive manual modelling or expert knowledge (Di Francescomarino et al. 2018; Friederich et al. 2022; Van Der Aalst 2018). Finally, process mining provides a good basis for analysing material flows and uncovering bottlenecks, batching and delays through a performance spectrum visualisation (Denisov, Fahland, and Van Der Aalst 2018).

Process mining has been applied in a variety of domains, such as healthcare, finance, education, manufacturing, or logistics (Garcia et al. 2019). Recent developments in the field have extended the capabilities of traditional process mining, allowing it to be applied on the data sources, that are typically available along supply chains, thus enabling it to improve SCM (Van Der Aalst 2019, 2021; Van Zelst et al. 2021).

In this study we analyse the potential of process mining for applications in SCM. We review recent technological advances in the process mining community that enable its application in the SCM domain and describe seven relevant techniques. Next, we combine the identified methodologies with expert interviews, to derive use cases where process mining can significantly contribute to supply chain efficiency and resilience. Each use case is cross-validated through further expert interviews and described in detail, demonstrating how process mining can provide actionable insights for a wide range of supply chain problems. We discuss the implications of our findings for both practitioners, who benefit from improved visibility and optimisation opportunities, and researchers, who are provided with a roadmap for further exploration of this promising interdisciplinary field.

In our study, we find that for practitioners, process mining is applicable in a wide range of use cases, revealing

the actual behaviour of supply chain operations. Furthermore, we find that process mining can be a valuable addition to the toolset of SCM researchers, adding a new perspective that fits well with the structure of supply chains. It can be the bridge to test and verify theory that has been built using manual modelling and simulation by mining large amounts of actual data. However, process mining in the SCM domain needs to be approached with caution and we find a number of limitations that need to be considered when using this technique as a basis for operational decision making.

To the best of our knowledge, this study is the first to explore specific use cases of process mining technology in a supply chain context. We contribute to existing operations research by integrating this mature technology into the broader landscape of SCM phenomena and established methodologies. This study also serves as a foundational resource for future research, encouraging scholars to explore the potential of process mining as a complementary tool in the SCM domain.

The remainder of this paper is organised as follows. Section 2 provides a background on traditional and advanced process mining methods. Section 3 describes our research methodology. In Section 4, we present our results and give a detailed description of each final use case. In Section 5, we discuss practical implications, limitations and give an outlook for further research. We conclude in Section 6 by summarising major research outcomes of our study.

2. Process mining background

2.1. Traditional process mining

Process mining emerged in the early 2000s at the intersection of data science and business process management. It aims to derive insights from so-called event log data stored in information systems within organisations (Van Der Aalst 2016). The field has continuously evolved, driven by improvements in data storage capabilities, computing power, and sophisticated algorithms that enable the analyses of large data sources.

The core idea of process mining is to automatically generate models that accurately represent how business processes are actually executed, as opposed to how they are expected to be executed (Van Der Aalst et al. 2012). It overcomes the limitations of traditional business process management tools by enabling the analysis of data readily available in information systems that reflect past instances of business operations (Garcia et al. 2019). This provides the basis for dynamic process mapping and subsequent improvement, leading to more efficient processes.

The foundation for process mining is event log data. Such event logs capture how historical instances of a process were executed. Table 1 shows a simple example of an event log for a make-to-order process. Each row in the event log must contain at least three required attributes: (i) an activity name that indicates the specific event that occurred (e.g. Create Production Order); (ii) a case identifier that links the event to a specific execution of the process (e.g. Order1); and (iii) a timestamp that indicates when the event occurred (e.g. on May 1, 2023 at 11:25am).

An event log may include additional process-related attributes such as (iv) an event identifier that uniquely references each event, i.e. each row (e.g. e1); (v) resources that remain constant throughout the entire case, such as material (e.g. M1); (vi) resources associated with individual activities, such as users (e.g. user1).

A central task in process mining is *process discovery*, which involves generating models from event logs that accurately represent the actual execution of past process instances (rather than of relying on documented to-be process descriptions). A key challenge is to derive models that avoid overfitting, i.e. being too detailed and capturing rare outliers as if they were frequent events, and

underfitting, i.e. being too general and omitting essential details that are crucial for an accurate process representation (Van Der Aalst et al. 2010).

A variety of mining algorithms have been proposed for process discovery, each with different strengths and weaknesses (Garcia et al. 2019). All algorithms aim to derive a process model, traditionally a Petri net, that best describes the process behaviour stored in the event log. In addition to Petri nets, these process models can also be represented in various other representations, also called notations, such as 'Business Process Model and Notation' (BPMN) or a directly follows graph as shown in Figure 1, which is the output format supported by many tools.

In addition to process discovery, there are three other types of traditional process mining: conformance checking, process reengineering and operational support (Van Der Aalst 2019). *Conformance checking* involves comparing actual process executions recorded in event logs with an expected, so-called 'normative' process model that can be manually generated or mined through process discovery. This comparison helps to identify deviations, compliance issues and inefficiencies, and provides a measure of how closely reality matches the desired process flow.

Table 1. Example event log data of a make-to-order process (required attributes are ActivityID/Activity, CaseID and Timestamp).

EventID	ActivityID/Activity	CaseID	Timestamp	MaterialID	UserID	...
e1	CCO – Create Customer Order	order1	01-05 09:05	M1	user1	...
e2	CPO – Create Production Order	order1	01-05 09:30	M1	user1	...
e3	CI – Create Invoice	order1	01-05 11:25	M1	user2	...
e4	RPO – Receive Production Order	order1	02-05 11:55	M1	user3	...
e5	RP – Receive Payment	order1	03-05 16:15	M1	user2	...
e6	PS – Pack Shipment	order1	03-05 16:30	M1	user3	...
e7	S – Ship	order1	05-05 11:00	M1	user3	...
e8	CI – Clear Invoice	order1	06-05 10:30	M1	user2	...
e9	CCO – Create Customer Order	order2	06-05 16:05	M2, M3	user1	...
e10	CPO – Create Production Order	order2	07-05 09:45	M2, M3	user1	...
e11	RPO – Receive Production Order	order2	07-05 15:25	M2, M3	user3	...
e12	CI – Create Invoice	order2	07-05 16:00	M2, M3	user2	...
e13	RP – Receive Payment	order2	07-05 16:55	M2, M3	user2	...
e14	PS – Pack Shipment	order2	07-05 17:00	M2, M3	user3	...
e15	S – Ship	order2	07-05 16:15	M2, M3	user3	...
e16	CI – Clear Invoice	order2	07-05 10:00	M2, M3	user2	...
...	...	...	...	...	...	...

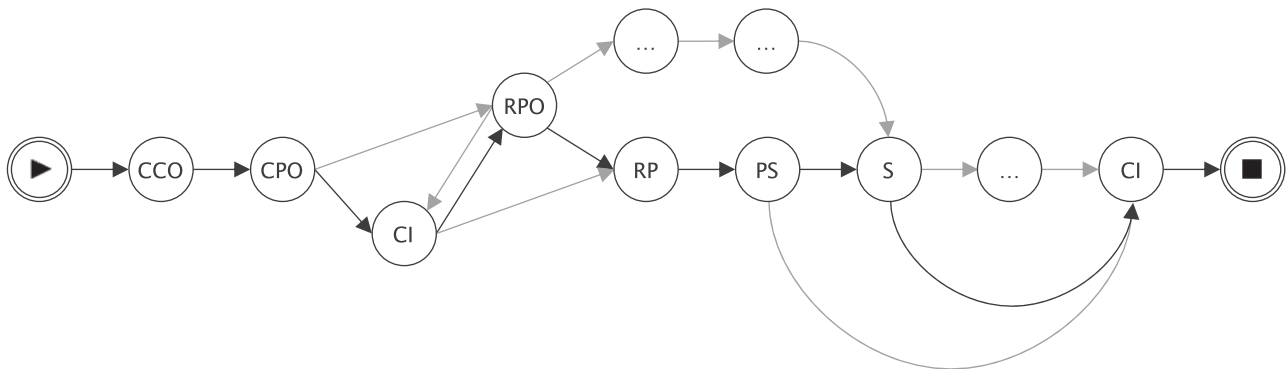


Figure 1. Process model of the make-to-order process in Table 1 as directly-follows graph.

Table 2. Existing SCM related and unrelated process mining use cases.

Domain	SCM-related	Example use cases
Healthcare	No	• Analysis of end-to-end patient journey in clinical processes (Partington et al. 2015) • Examining efficiency of hospital operations before and after construction of a new building (Yoo et al. 2016)
Finance	No	• Automating and accelerating financial statement audits (Werner, Wiese, and Maas 2021) • Financial fraud detection and mitigation (Jans et al. 2011)
Education	No	• Analyzing patterns and strategies in students' self-regulated learning (Bannert, Reimann, and Sonnenberg 2014) • Mining students' path through curriculum (Wang and Zaiane 2015)
Manufacturing	Yes	• Improving productivity in make-to-stock manufacturing (Lorenz et al. 2021) • Manufacturing completion time prediction (Choueiri et al. 2020) • Manufacturing quality assurance (Ho et al. 2009; Lee et al. 2014) • Detect and manage manufacturing system failures (Sánchez, Exposito, and Aguilar 2020)
Logistics	Yes	• Analysis of production support processes (Lugaresi et al. 2023) • Mined value stream mapping for internal logistics (Knoll, Reinhart, and Prügmeier 2019) • Visual identification of performance patterns in baggage handling process (Denisov, Fahland, and Van Der Aalst 2018)

Process reengineering aims to optimise an existing process model based on the insights gained from mining actual process behaviour from event logs. It aims to improve process performance to meet key performance indicators and optimise overall efficiency. Identifying the root causes of deviations and inefficiencies in the input model serves as the basis for eliminating bottlenecks and redundant activities that do not add value.

Operational support uses process mining to aid in the coordination and execution of ongoing process instances. Deviations from the intended path are identified in real time and communicated through warnings, predictions or recommendations.

2.2. Existing process mining applications

Process mining has been applied in a wide range of domains. Table 2 lists applications in five different domains, demonstrating how process mining is used to analyse and improve processes in domains related and unrelated to SCM.

Applications of process mining have traditionally focused on service industries with little connection to SCM. In healthcare, process mining has been used to improve patient flow and optimise treatment processes (Partington et al. 2015; Yoo et al. 2016). In the financial sector, process mining opens up automated ways to conduct financial statement audits and detect fraud (Jans et al. 2011; Werner, Wiese, and Maas 2021). In the education sector, process mining has been applied to map students' online self-study processes and their journeys through the entire curriculum (Bannert, Reimann, and Sonnenberg 2014; Wang and Zaiane 2015).

More recently, studies have explored the application of process mining in domains that are inherently related to SCM. In manufacturing, process mining is used to map the material flow of production lines, predict completion times, and analyse production support processes (Choueiri et al. 2020; Lorenz et al. 2021; Lugaresi et al. 2023). Other studies investigate process mining as a

tool for production quality assurance and system failure detection (Ho et al. 2009; Lee et al. 2014; Sánchez, Exposito, and Aguilar 2020). In logistics, process mining is used to perform value stream mapping of internal processes and to measure system performance (Denisov, Fahland, and Van Der Aalst 2018; Knoll, Reinhart, and Prügmeier 2019).

2.3. Advanced process mining methods

Recently, advanced techniques, summarised in Figure 2, have emerged in the process mining community that extend the capabilities of traditional process mining and make it applicable to large event log data sources, such as the operational systems found along supply chains. These advances include techniques for processing event logs so that they can be mined across multiple operational information systems and handle the flow of individual objects (Sections 2.3.1. to 2.3.3.), and techniques for improved analysis, simulation and prediction of mined process models (Sections 2.3.4. to 2.3.6.).

2.3.1. Object-centric process mining

Traditional process mining represents processes as multiple cases (e.g. customer orders) that flow through sequences of activities (e.g. create customer order, receive payment, clear invoice). However, for processes involving multiple objects (e.g. different materials in manufacturing), this can create an inaccurate picture. Object-centric process mining addresses the complexity of processes that involve multiple interacting objects. This is a common scenario in operations and SCM. In contrast to traditional process mining approaches, object-centric process mining utilises event logs that reference multiple objects, such as orders, products, suppliers and shipments (Van Der Aalst 2019). This technique allows for the representation of interactions between these objects, providing a more comprehensive and accurate understanding of the processes involved.

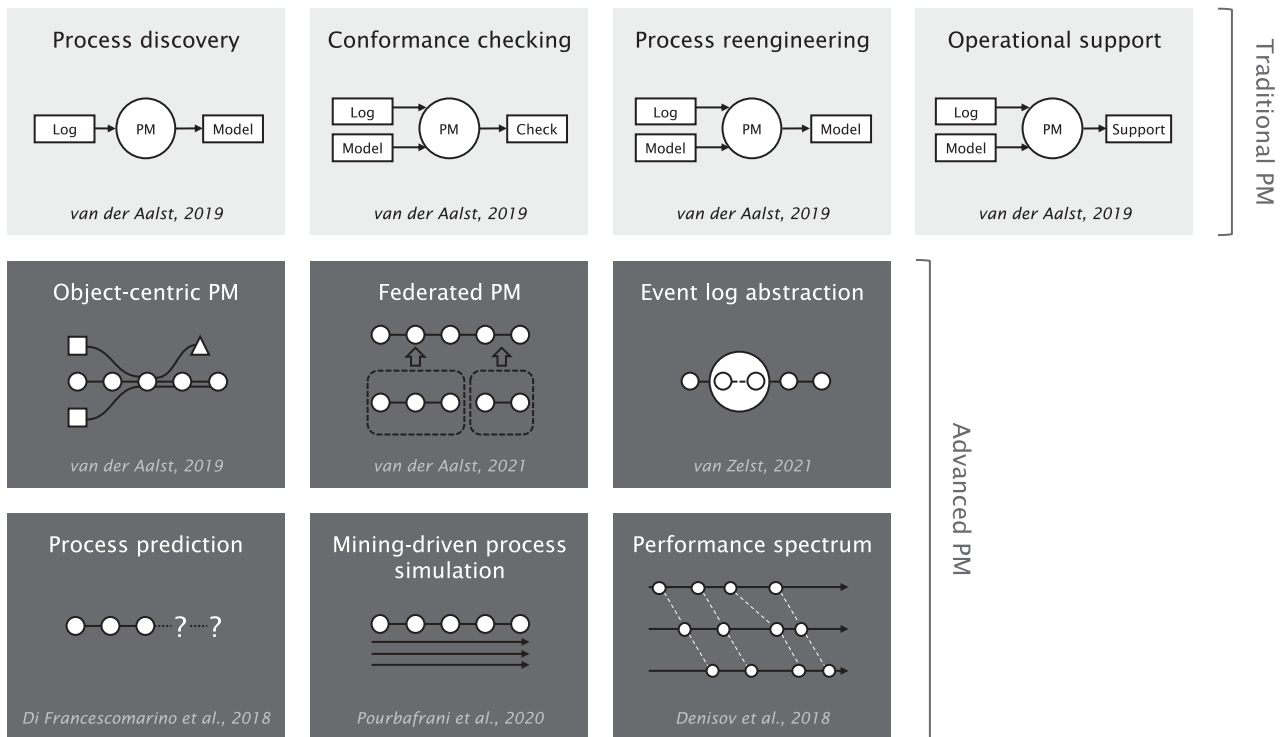


Figure 2. Traditional and advanced process mining techniques.

Object-centric process mining effectively represents the convergence and divergence of objects. For example, in manufacturing or assembly processes, convergence could involve the combination of raw materials or components into a finished product, while divergence could refer to waste products, such as leftover material from a cutting activity. In the context of SCM, this approach helps to uncover bottlenecks, inefficiencies or opportunities for optimisation by creating transparency of how objects such as orders, materials, or workers interact. These are insights that remain hidden in traditional case-centric process mining.

2.3.2. Federated process mining

Traditionally, process mining has focused on analysing of isolated processes recorded in event logs from a single data source. However, in many real-world scenarios, process-related data is distributed across multiple systems, each controlled by different organisational units or companies. Unifying information from distributed data sources poses certain challenges in terms of competing interests and privacy issues, as well as technical complexities in combining event logs from different system architectures. Federated process mining addresses these challenges by introducing two ways to generate a global process model across multiple systems (Van Der Aalst 2021). The first way assumes the willingness of partners to share process-related event logs. In this case, the

challenges are limited to technical complications such as merging identifiers and bringing event logs to the same level of granularity.

The second way enables organisations to share high-level process information by mining local process models and sharing abstractions of them. The resulting global model in this case is similar to multi-echelon models used in operations and supply chain research to model buffer stock across local inventories (Van Der Aalst 2021). Federated process mining extends the scope of possible analyses to processes that span multiple systems or even companies. An end-to-end view of operations and supply chain processes is usually only possible by incorporating multiple sources of collected event log data. The two proposed methods for achieving global process models allow both a detailed view of internal operations or logistics processes as well as a more holistic view of interactions such as material flows between companies, sharing only the minimum relevant information without compromising on data privacy and security.

2.3.3. Event abstraction

Event log abstraction in process mining focuses on refining and simplifying event logs to better match the conceptual level of analysis, thereby improving the interpretability and applicability of process mining techniques. This approach is critical for operations and SCM, where data from disparate systems often varies

in granularity and level of abstraction. By abstracting event logs, essentially grouping together low-level events, organisations can gain a higher-level understanding of their end-to-end processes. This in turn facilitates the identification of inefficiencies, bottlenecks and opportunities for optimisation across complex operational and supply chain activities. The abstraction process enables a clearer visualisation of process flows, making it easier to align operational performance with strategic objectives (Van Zelst et al. 2021). Furthermore, in the supply chain context, where processes often span multiple systems and organisational boundaries, event log abstraction supports the integration of disparate data sources through federated process mining (described in Section 2.2.2. above), providing a unified view of supply chain operations. Instead of sharing granular data, companies can collaborate by sharing abstractions where sensitive information is hidden.

2.3.4. Predictive process monitoring

The field of process mining is not only concerned with the inherent analyses possible through the technology itself but also with the interfaces to other data science disciplines such as machine learning. Research at the interface of process mining and machine learning is concerned with predicting the future of ongoing process instances. Known as predictive process monitoring, this encompasses three types of predictive approaches that can be conducted (Di Francescomarino et al. 2018): Numeric predictions for estimating variables like process cost or completion time (Van Der Aalst, Schonenberg, and Song 2011), categorical predictions usually related to binary outcomes such as customer churn (Teinemaa et al. 2018), and activity sequence predictions for forecasting the future activities within an ongoing case (Polato et al. 2018). The first two types can be addressed using traditional machine learning methods based on tabular descriptive data. However, encoding process traces into a set of variables that machine learning models can be trained on may increase the predictive performance. The third type is unique to the domain of process mining and opens up completely new ways of predicting deviations of ongoing process instances.

2.3.5. Mining-driven process simulation

Simulation models have traditionally been used in SCM research to mimic real-world scenarios. The effectiveness of these simulations is highly dependent on the accuracy of the underlying assumptions or parameters, such as the definition of events, possible paths and their associated probability distributions. Manually configuring these parameters is a resource-intensive task. Furthermore, the assumptions made are usually static and run

the risk of losing their validity in dynamically changing environments (Karlsson 2016). Through process mining, the creation of such simulation models in SCM can be simplified by extracting models directly from data. This accelerates the initial workload of setting accurate parameters and conducting simulation.

Discrete event simulation involves defining rules to generate events that change the state of the system at different times, allowing different ‘what if’ scenarios to be explored through simulation experiments. However, building a robust simulation model is often time-consuming and can quickly become outdated, limiting its practical application. In contrast, process mining starts with observed behaviour, using event data to detect deviations and analyse performance issues. It generates process models that can then be used as the basis for simulation, increasing the relevance and timeliness of the model.

The scope of such mined models is chosen either to simulate specific instances through agent-based simulation or to view the process from a higher level of abstraction, a so-called system dynamics approach, where simulation is used to predict the effect of process improvement actions (Pourbafrani, Van Zelst, and Van Der Aalst 2020). Thus, by automating the creation of simulation-ready models, process mining not only eliminates the need for manual assumptions and domain knowledge, but also ensures that these models remain current and based on actual data, enabling a more agile response to changing real-world conditions.

2.3.6. Process performance analysis

Visualisation methods in process mining usually involve the aggregation of case instances to reveal the process flow and sequence of activities shared by the majority of cases. However, in this view important information related to the temporal patterns of the cases remains hidden. Process performance analysis, specifically through a performance spectrum, provides an alternative visualisation method to evaluate process behaviour and its efficiency over time without aggregating event data. By mapping each step of a process over time, this technique reveals the variability and synchronisation of case performance, revealing patterns that may be obscured by traditional process visualisation methods (Denisov, Fahland, and Van Der Aalst 2018). The performance spectrum facilitates a detailed examination of process flows, allowing the identification of both expected and unusual performance patterns. It significantly improves the understanding of non-stationary process behaviour and inter-case dependencies, providing valuable insights for operations and SCM. The granular view of this method helps to uncover bottlenecks,

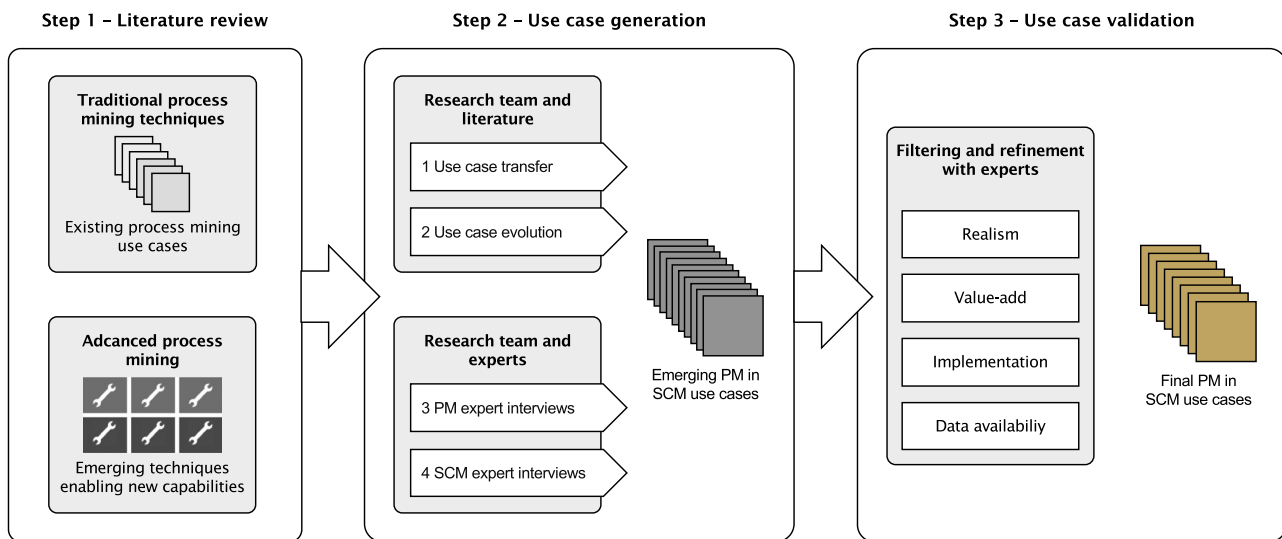


Figure 3. Research methodology for collecting and validating use cases.

understand resource utilisation and identify opportunities for process optimisation, ultimately leading to more informed decision making and strategic planning in supply chain and operational contexts.

3. Methodology

Our research aims to identify appropriate use cases for process mining technology in SCM, demonstrating its potential enabled by emerging advanced process mining techniques. Figure 3 outlines our research approach. As a first step, we review the relevant literature. Here, we considered literature on traditional process mining techniques applied in domains both related and unrelated to SCM (see Section 2.2). In addition, we reviewed recent advances in the process mining field that enable its application in the SCM domain (see Section 2.3).

In the second step of our research, we follow a dual approach to collect use cases through prior literature and expert interviews. The research team analysed existing use cases reported in the literature for their applicability to the supply chain domain. First, we consider existing use cases from SCM-unrelated domains, such as finance, that could be transferred to the SCM domain. Second, we consider use cases from SCM-related domains, such as manufacturing, that could be scaled using advanced process mining techniques to fit the larger SCM domain, where material and information flow through multiple entities and are stored in federated systems.

In addition, we involved experts in the generation of use cases. While process mining is traditionally an academic discipline, much of the current development in practice is taking place within process mining software developers or specialised departments within companies.

Consequently, many of the relevant use cases for SCM are quite new and therefore not yet reported in the academic literature. In order to include use cases that have been developed in practice, we supplement our collection process with expert interviews. We conducted 21 initial interviews: 15 with supply chain experts familiar with process mining technology and its use, and 6 with representatives of process mining software solution developers, ideally with knowledge and experience in the SCM domain. A complete overview of all expert interviews is presented in Appendix 1. The experts included in our interviews are senior executives from major industrial companies including the chemical, automotive and semiconductor industries, among others, who manage global manufacturing supply chains, as well as the leading developers of process mining solution providers. We approached key contact persons from previous research projects, academic contacts within software companies, supplemented by a LinkedIn search using the keywords 'process mining' in combination with 'supply chain management' and 'supply chain planning'. After the initial contact, we were often referred internally to the experts who best fit our research topic at the intersection of process mining and SCM. It was important for the selection of interviewees that the applications of process mining in their work environment went beyond standard use cases such as the analysis of order-to-cash (OTC) and procure-to-pay (PTP) processes, which have a fast path from implementation to value creation and are therefore currently the primary applications of process mining in practice, but are not interesting from the perspective of SCM research.

In a first round of semi-structured interviews, the research team first introduced the objective of the

study and, depending on the interviewee's background, provided an overview of the emerging capabilities and recent advances in process mining technology (as outlined in Section 2.2.). In a discussion with the experts, we then went through existing problems within their company and department and iteratively tested the potential application and benefits of process mining against existing analysis tools currently in use. The most realistic and promising results from these interviews were added to the collection of initial use cases.

After collecting eleven initial use cases, we moved on to the third step of our research to validate these use cases. Here, we conducted a second round of structured interviews with five experts from the first round of interviews to validate our initial findings. These interview partners were selected to best match experts from backgrounds and industries that were different from the perspective of the initial interview or literature source, to ensure the generalisability of our findings. Each of the initial use cases is discussed with at least two experts. In these interviews, we specifically validate four elements: First, we obtain the expert's opinion on the overall realism of the use case within their specific area of expertise. Second, we ask the expert to validate the added value described by the research team. Third, we ask the expert to outline the challenges and limitations in implementing the use case. Fourth, the expert explains whether the use case would be feasible with the data collected in their own systems and what system modifications or data transformations would be required.

Through this process, we filter and refine the use cases based on expert input. In addition to minor modifications to most of the use cases, we discard two use cases altogether after failing validation with respect to general data availability, discard two use cases after failing validation with respect to data sharing and sovereignty issues, and merge two other use cases into one because they were closely related and more relevant when combined. The resulting set of 6 final use cases is presented in detail in the following section.

4. Results

We report six use cases that demonstrate the potential of process mining technology across a wide range of SCM applications. The use cases are summarised in Table 3. For each use case, we provide a description of the applicable traditional and advanced process mining concepts, outline the advantages of the PM approach over traditional approaches, and the generation method as described in Section 3. Each use case is then described in detail in the following subsections.

4.1. Use case 1: planer workflow discovery

Many supply chain planning processes are increasingly adopting automation. However, in most companies, SC planning is still a mix of automated and manual processes, or semi-automated processes, that drive the best outcomes. Particularly in complex setups, crisis situations, or during the transition from manual to more automated systems, the intervention of a human planner can add substantial value (Fildes et al. 2009; Perera et al. 2019). However, quantifying this value and identifying relevant activities in the planning process, however, poses its own set of challenges. Process mining provides a comprehensive solution for this issue, as all activities of the planner that are recorded in SC planning platforms can be traced and analysed.

Consider a semi-automated inventory management system that can suggest restocking decisions based on various parameters (such as supplier lead time, batch sizes or reorder points), current inventory levels, and predicted demand. The role of the planner then becomes critical in assessing these proposals, especially considering factors that the system may overlook, such as upcoming promotions, supplier delays or recent market trends. Planners can validate, modify or overrule the system's suggestions based on their expertise (Imdahl, Schmidt, and Hoberg 2022) at the moment of truth, or adjust planning parameters that will alter future decisions.

By tracking all interventions along the entire decision-making process, process mining can compare the initial system proposals with the final decisions post-planner interventions. Traditional approaches compare only the final outcome, such as forecast adjustments (Khosrowabadi, Hoberg, and Imdahl 2022). With process mining, however, the planner's entire decision-making process becomes transparent, and value-adding activities can be separated from non-value-adding activities.

The application of process mining in discovering planner workflows is further enriched by its interface with task mining. Task mining provides a granular perspective on the activities and workloads of individual planners or teams, analysing the specific tasks, applications and actions used to achieve planning objectives. This detailed analysis of task-level digital footprint data across systems and tools complements the process-level insights provided by process mining, providing a more nuanced understanding of the planning process (Choy, Gazzo, and Panikkar 2024).

4.2. Use case 2: disruption impact discovery

SCM is inherently exposed to various uncertainties, making the management of supply chain disruptions a critical aspect of maintaining operational continuity

Table 3. Selected use cases for process mining in SCM.

#	Use case	Description	Advantages over traditional methods	Applicable process mining concepts	Generation method and source	Limitations and considerations	Validation
1	Planer workflow discovery	Trace the decision-making workflow of supply chain planners. Discover negative workarounds that limit planning efficiency, positive workarounds that increase efficiency, and interventions in automated planning systems.	Objective, data-driven tracking overcomes the limitations of the planner's subjective view that would influence traditional methods such as planner observation, surveys or interviews.	Process discovery, process reengineering	Interview #2 with supply chain expert	Planning may be (partially) manual and not recorded in structured systems (paper-based and Excel sheets); issues related to privacy of planner data	Validated in interview #22 with supply chain expert; validated in interview #25 with process mining expert
2	Disruption impact discovery	Analyze the impact of an upstream disruptive event on downstream levels of the supply chain. Quantify at network level between collaborating entities and at object level by tracking individual material flows through the manufacturing process.	PM enables the quantification of the impact of disruptions at a granular level, allowing the analysis of individual material flows far upstream across multiple collaborating companies to identify the impact of incoming disruptions earlier. Traditionally, impact is measured either at a local level (e.g. stock-outs leading to increased lead times) or at a very high level, such as the impact of disruption on business valuation.	Process discovery, object-centric PM, federated PM, event abstraction, performance spectrum	Interview #14 with process mining expert	Identifying temporal patterns is only possible in high-volume, low-variety supply chains (optimally make-to-stock); following individual material flows along the SC requires a high level of data sharing between partners.	Validated in interview #23 with supply chain expert; validated in interview #26 with supply chain expert
3	Value stream mining and lean process optimisation	Overcome the limitations of manual VSM by mining production process steps from data. Enables accurate, dynamic, and automated VSM for comparison with existing maps and operational process enhancement.	Traditional VSM relies heavily on manual work and domain knowledge, making it slow and resource intensive. Process-data-based value stream mapping is faster and provides an objective view of material flows.	Process discovery, conformance checking, process reengineering, operational support federated PM, event abstraction	Evolution of existing research on value stream mining in internal logistics by Knoll et al., (2019)	Requires system landscape capable of extracting end-to-end material flow data; difficult in production environments that are not discrete (e.g. chemical industry) because quantity measures such as weight between production steps are not tracked in the data; limited research exists on converting Petri nets to VSM representation.	Validated in interview #22 with supply chain expert; validated in interview #25 with supply chain expert
4	Regulation and compliance adherence	Ensure supply chain compliance by mining relevant sub-processes. Share abstractions between supply chain partners as proof of compliance without sacrificing data sovereignty.	Traditionally subject to audit by external certification providers based on samples. Process mining enables a more complete view of all production iterations, leading to greater confidence in production quality (potentially even in real time or for filtered materials & orders).	Process discovery, conformance checking, event abstraction	Interview #8 with supply chain expert	External certification is still required but would need to move towards assuring the correctness of the data collection process rather than samples of the manufacturing process itself; medium and format for sharing process abstractions is not currently standardised and will need to be customised.	Validated in interview #22 with supply chain expert; validated in #23 with process mining expert
5	Multi-echelon inventory model mining	Mine inventory models across operational systems to discover bottlenecks and inefficiencies in reordering logic. Simulate alternative parameters to optimise performance across the supply chain.	Multi-echelon models and simulation exist and are a commonly used technique to optimise inventory parameters. Process mining can increase accuracy by mining the model directly from data and adaptability to frequently changing situations, taking into account upstream and downstream inventory levels.	Process discovery, process reengineering, operational support, federated PM, process simulation	Interview #10 with supply chain expert; also discussed in existing research on federated process mining by Van Der Aalst (2021)	Coordination of multiple supply chain tiers to mine process abstractions that can be combined.	Validated in interview #23 with process mining expert; validated in interview #25 with supply chain expert
6	End-to-end network visibility	Identify inefficiencies in cross-entity manufacturing that limit supply chain performance as measured by lead times. Detailed tracking of individual cases and objects to mitigate inefficiencies in ongoing cases.	Supply chain performance is measured by defined KPIs, such as lead time. While these KPIs can be accurately tracked in standard systems, understanding changes, such as lead times, and identifying opportunities for optimisation is where process mining can improve the current state.	Process discovery, object-centric PM, federated PM	Interview #14 with supply chain expert	Underlying data systems can be highly customised; detailed data sharing is required to analyse the supply network outside an organisation.	Validated in interview #24 with supply chain expert; validated in interview #26 with supply chain expert

and resilience. While empirical studies have quantified the impact of disruptions at an abstract level, such as by analysing the stock price performance of interconnected manufacturing firms, they lack granular information about the underlying material flow between firms (Hosseini, Ivanov, and Dolgui 2019; Munoz and Dunbar 2015). In addition, network-based modelling methods have been used to study the propagation of disruptions in a supply chain network, but lack a connection to actual data sources (Katsaliaki, Galetsi, and Kumar 2022; Kim, Chen, and Linderman 2015). Process mining combines data mining and modelling to understand the full scope of disruptions by capturing end-to-end material flows, thereby enabling a detailed analysis of the impact of disruptions on downstream tiers of the supply chain.

Consider the scenario of extended port downtime due to a disruptive event, such as the Tianjin port explosion in 2015 (Fu, Wang, and Yan 2016). Traditional analytics can quantify the immediate damage and local supply chain impacts, such as stock outs and increased lead times, but cannot directly link these local impacts back to the initiating disruption. Analysing material flows through process mining across multiple tiers of the supply chain provides visibility into previously intangible connections between downstream supply chain KPI deviations and the original disruption event. This includes detailing how shipments were delayed and rerouted, how production orders requiring that material were postponed, and how assembly requiring components were shifted to alternative suppliers.

In the field of SCM disruption management, process mining not only illustrates the immediate impact within the supply chain but can also be used to reveal the ripple effect, as discussed by (Dolgui, Ivanov, and Sokolov 2018). The ripple effect describes how a single disruption can cascade through the supply chain, affecting nodes far downstream from the initial event. For example, a delay in the delivery of raw materials due to a disaster-affected area can later cause production bottlenecks, leading to delivery delays and increased inventory costs even in regions unaffected by the initial disruption. By linking specific disruptive events to anomalies within the supply chain through process mining, managers can visualise and quantify both immediate and cascading effects, enabling better preparedness and more robust supply chain designs.

The impact of process mining in analysing supply chain disruptions extends to two main levels: the network level and the transaction level. At the network level, process mining can reveal changes to alternative suppliers, transport routes or modes, providing strategic insight into how companies respond to different disruptive events. At the transactional level, individual objects

can be traced from the point of disruption through the supply chain, identifying specific points of failure such as stock outs or manufacturing line delays. This dual-level analysis could have new implications for supply chain resilience strategies that not only address the local impact of disruptions, but also mitigate the propagation of these impacts throughout the supply chain.

4.3. Use case 3: value stream mining and lean process optimisation

Value stream mapping (VSM), a fundamental method of lean management, traditionally involves manually tracing the flow of materials in a pen-and-paper fashion from raw materials through various manufacturing stages to the final product (Rother and Shook 2018). While effective in identifying inefficiencies and optimising process flows, this method requires significant manual effort and deep domain knowledge, making it time and resource intensive to use on a regular basis (Gupta, Modgil, and Gunasekaran 2020).

Process mining could address the challenges of traditional VSM by providing an efficient way to mine value stream maps directly from event log data stored in information systems, i.e. digital traces of the production process in action. Advanced process mining allows information stored across multiple production systems or entities to be linked, providing a comprehensive automated baseline for VSM. This significantly reduces manual effort and enables more frequent analysis.

Consider a make-to-order manufacturing environment where VSM is used to streamline the production process and eliminate waste. Here, process mining can analyse historical data containing production information from completed orders to identify the most common material flows. It facilitates the identification of bottlenecks, unnecessary steps or unwanted deviations from the intended production process, thereby contributing to lean process improvement.

This approach not only complements traditional VSM by highlighting material and information flows as they actually occur, but also reveals discrepancies between perceived and actual process flows. Companies can use existing value stream maps that represent how they expect the production process to look like, and apply process mining to check in how far the actual production conforms with this ideal.

4.4. Use case 4: regulation and compliance adherence

Ensuring adherence to regulations and compliance standards is a central aspect of modern business operations,

especially in industries where regulatory requirements dictate operational processes (Manavalan and Jayakrishna 2019). The challenge of compliance lies not only in understanding and interpreting complex regulations, but also in ensuring that every step of the operational process aligns with these regulations. Traditional methods of ensuring compliance, such as random sampling, may not provide the comprehensive oversight required to ensure full compliance across all operations.

Process mining provides a robust solution to these challenges by enabling end-to-end analysis of operational processes against regulatory requirements and compliance standards. For example, in industries subject to the Supply Chain Due Diligence Act, companies are required to implement due diligence obligations within their supply chains to prevent human rights violations and environmental damage. Process mining can play a critical role in ensuring compliance with such regulations by providing visibility into the entire supply chain, from the sourcing of raw material to the final delivery of products.

Another example is a scenario where quality assurance in a manufacturing company is not only a marker of product excellence, but also a regulatory requirement. Traditional quality assurance methods may involve randomly sampling products to check for compliance with pre-defined standards. However, this approach may not capture all instances of non-compliance and can lead to regulatory risks.

By applying process mining, an organisation can analyse the complete set of event logs from production systems to identify any deviations from the required quality assurance processes. It can identify systematic issues that lead to non-compliance, such as machine breakdowns, operator errors, or inconsistencies in raw material quality. Unlike random sampling, which provides a snapshot in time, process mining provides a continuous check of compliance adherence, ensuring that each product meets the required standards and regulations.

In addition, companies can share abstracted versions of their process models as evidence of regulatory compliance. This approach involves abstracting relevant parts of internally mined process models into simplified models that highlight compliance-related activities without disclosing sensitive business information. By sharing these abstractions, organisations can demonstrate proof of compliance to regulators and partners in a manner in an efficient and secure manner.

4.5. Use case 5: multi-echelon inventory model mining

The integration of operational data to build multi-echelon inventory models holds potential for SCM

research, adding a new analytical dimension to the discussion of supply chain phenomena. Phenomena such as the bullwhip effect, where small fluctuations in demand at the retail level cause increasingly larger fluctuations up the supply chain, have long affected supply chains, leading to inefficiencies such as overstocking or stock-outs. Research on the bullwhip effect has traditionally been based on either simulation models that rely on manual assumptions or aggregate data analyses that do not reflect information and material flows, partly due to the complexity of sharing data across different supply chain echelons and organisations (Chatfield 2013; Wangphanich, Kara, and Kayis 2010).

Federated process mining offers a novel approach to building dynamic, data-driven, multi-echelon models. This technique enables the mining of process models directly from real data, providing an accurate representation of inventory flows and interactions across multiple supply chain tiers. In this context, federated process mining proposes a solution for how organisations can collaborate to share abstractions of their operational data while maintaining data security and sovereignty, overcoming a key barrier to effective supply chain optimisation.

This shared data enables the construction of comprehensive multi-echelon models that accurately reflect the flow of information and materials along the supply chain. This mined model can be simulated to test alternative scenarios for parameters such as reorder points or batch sizes. The goal is to find efficient solutions that optimise not only the operations of a local system, but all the interconnected entities.

This approach also enables continuous, real-time analysis of supply chain dynamics, allowing companies to adapt their strategies as conditions change. For example, a sudden increase in demand at the retail level can be quickly communicated up the supply chain, enabling upstream suppliers to adjust their production and inventory levels accordingly, reducing the risk of overstocking or stock-outs.

4.6. Use case 6: end-to-end network visibility

End-to-end lead time tracking in SCM is essential for understanding the total time it takes from the moment an order is placed until it is delivered to the customer. This metric is critical for identifying areas of inefficiency, predicting accurate delivery times and ultimately, improving customer satisfaction. However, accurately tracking this metric across a complex supply chain that may involve multiple manufacturing sites, suppliers and modes of transport is inherently challenging. Traditional approaches to measuring end-to-end lead times often rely on manual data collection and aggregation, which is

time-consuming and prone to error, resulting in a lack of real-time visibility and actionable insights (Sodhi and Tang 2019).

Process mining opens up analytics beyond calculating lead time as the delta between order placement and delivery. From event log data, it becomes possible to trace historical orders end-to-end through each step of the value-adding process. Process mining algorithms add transparency to how orders are handled and how individual objects such as materials, people or documents interact with each other. This added process visibility can provide the basis for identifying inefficiencies and reduce lead times.

Consider a scenario where a company wants to reduce its overall delivery times to improve customer satisfaction: process mining can provide detailed insights into each step of the supply chain process. For example, it can uncover hidden bottlenecks such as delays in supplier response, production inefficiencies or transportation issues that contribute to extended lead times. By pinpointing the specific stages where delays occur, companies can implement targeted improvements, such as optimising production scheduling, renegotiating supplier contracts, or exploring alternative transportation strategies.

In addition, process mining enables continuous monitoring of lead times, allowing companies to track the impact of implemented changes in real time. This continuous feedback loop is essential for sustaining improvements and dynamically adapting to changes in the supply chain. For example, when a new supplier is onboarded or a new transport route is introduced, companies can immediately assess the impact on overall lead times and make further adjustments as necessary.

5. Discussion

5.1. Managerial implications

The application of process mining to SCM offers significant benefits to practitioners by improving visibility, ensuring compliance and providing a basis for operational improvement. A supply chain consists of a large number of interconnected sub-processes that operate in a highly dynamic environment. By implementing process mining, practitioners gain an in-depth view of these sub-processes, enabling them to identify and analyse specific instances where inefficiencies are impacting operational effectiveness.

However, process mining has rarely been applied to SCM beyond a few off-the-shelf solutions that focus on optimising the order-to-cash or procure-to-pay cycles. Widespread SCM applications are still in their infancy

in most of the companies that we had the opportunity to sample. Advanced process mining techniques are enabling larger-scale analysis, broadening the focus to include overarching dynamics and flow patterns that operate across multiple production systems or even between collaborating entities. This capability enables a comprehensive understanding of the entire supply chain landscape, facilitating informed management decisions based on dynamic, process-data-driven insights, rather than relying solely on manual mapping or aggregate statistics. Our research indicates that in order to leverage PM techniques, companies need to embed process mining expertise more deeply within the SCM community.

Companies need to provide supply chain managers with a holistic understanding of the concepts and application opportunities of process mining. Only once provided with this knowledge, they can interact successfully with process mining experts to fully leverage and implement the technology. The use cases we have identified can certainly act as a starting point for inspiration and further application and show that many avenues might emerge. However, still further exploration and implementation is needed to take full advantage of PM. This situation, at least to a certain extent, resembles the use of machine learning and artificial intelligence for SCM some 10 or 15 years ago, when little was known about the use cases and the execution of these techniques in an SCM context.

5.2. Limitations

Our research aims to provide an overview of how process mining can be used in various areas along the supply chain. However, many more use cases than those identified in Section 4 exist. The use of process mining technology is an alternative method of gaining visibility into the process-driven supply chain domain, rather than just providing benefits for a defined set of use cases. Whether for isolated processes within a single department or end-to-end material flow across multiple manufacturing entities, analysing event log data holds great potential for identifying improvement opportunities.

The use of process mining as a basis for actual operational improvement in the SCM domain is subject to a number of challenges identified in our interviews with partners from different SCM areas.

First, process mining assumes that the event log data fed to the mining algorithm represents actual cases that have occurred in reality. However, event logs do not always accurately represent reality and must be carefully validated. A distinction must be made between data that is recorded in a fully automated manner, such as by sensors, and data that is recorded in a fully or partially manual fashion. Much of the event log data stored

in ERP systems is based on manual user input, such as recording the receipt of goods. In such cases, there is potential for input errors or inaccuracies in the completeness and timeliness of events. This must be taken into account when analysing the results of process mining and using them to inform operational and supply chain decisions.

Second, data sharing in cross-organizational settings remains a challenge. Consequently, the use of process mining to analyse end-to-end processes is subject not only to technical difficulties, such as matching case identifiers, but more importantly to data sovereignty considerations. The degree of willingness to collaborate and share data between connected companies in a supply chain may not be given. However, cross-entity process analysis has great potential in mitigating operational problems. The federated process mining approaches described in Section 2.2.2 allow companies to share abstractions of their process data that can be used to measure high-level patterns such as the bullwhip effect. However, such projects require a certain degree of co-operation between companies as well as a basic understanding of the aggregation techniques used.

Thirdly, our interviews with supply chain experts in various industries have shown that the quality of mining material flow processes depends on the underlying material. In discrete manufacturing, individual parts have individual identifiers that can be tracked in information systems. In other contexts, however, such as the chemical industry, this is not the case. Object-centric process mining is a theoretical solution for representing discrete objects, but in industries where material can only be measured in mass or weight, the capabilities of this method are currently limited.

5.3. Roadmap for future research

The primary pathway for using process mining in SCM research is through the use of process discovery. By visualising processes, researchers are able to gain transparency into large-scale interconnected events from operational systems. Concepts and phenomena discussed in SCM research can be enriched with a new perspective alongside existing methods.

The second interesting area for future research is the intersection between process mining and methods commonly used in the SCM community, such as machine learning and simulation. Here, process mining can act as a meaningful addition in providing process-context-related data to make existing models more accurate, or as a means to reduce the amount of manual effort involved in building models and setting parameters. Table 4 outlines potential research questions emerging from the six previously identified use cases.

6. Conclusion

Process mining technology enables organisations to gain visibility into how their processes actually run. It reveals deviations from the intended process, bottlenecks that limit process flow, and unnecessary activities or loops. While it has been successfully applied in various use cases from healthcare to manufacturing, its large-scale applicability to the SCM domain has been largely unexplored. Recent technological advances in the process mining community have proposed methods to connect data sources across multiple companies, represent individual objects and materials, change the level of abstraction, integrate simulation and machine learning techniques, and analyse process flow. In this study, we have reviewed

Table 4. Questions for future research emerging from each use case.

Nr	Use case	Questions for future research
1	Planer workflow discovery	Q1.1. What are typical points of intervention in semi-automated processes? Q1.2. Does planner intervention improve the planning process outcome? Q1.3. What is to learn from process deviations (positive/ negative workarounds) that planners have developed and deployed over time?
2	Disruption impact discovery	Q2.1. What are the impacts of supply chain disruptions on an aggregated network level (alternative suppliers, transport modes and routes)? Q2.2. What are the impacts of supply chain disruptions on item/ transaction level (actual traces of objects back to initial disruption explaining delays, stock-outs etc)?
3	Value stream mining and lean process optimisation	Q3.1. How do companies who conduct pen-and-paper VSM implement and trust process-mining-based VSM? Q3.2. Is fully automated VSM possible? What are limitations in terms of available data and under what conditions is human input necessary?
4	Regulation and compliance adherence	Q4.1. What are specific SCM regulations that can be covered by process mining? Are there scalable cases that apply to sector/ group of companies in the same way? Q4.2. How can process mining outputs be shared securely (federated process mining, role of a trusted third party and interfaces to blockchain/ zero-knowledge proof)?
5	Multi-echelon inventory model mining	Q5.1. How do automated batch sizes or reorder points affect downstream operations (optimal coordination of multiple systems)? Q5.2. What are the conditions that enable simulation using digital twin technology (constant material flow, fixed material paths)?
6	End-to-end network visibility	Q6.1. In how far compares process mining to existing methods of lead time tracking (advantages and disadvantages)? Q6.2. Does encoded process data improve existing lead time prediction methods?

these technological advances with regard to how they enable process mining for the analysis of SCM. Using a dual approach of collecting existing use cases in the literature and generating new ones in interviews with domain experts, we generate and validate six use cases that demonstrate the potential of process mining in the SCM domain. We discuss managerial implications for analysis in practice, directions for further research, and limitations of process mining in the specific SCM environment.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this study are available from the corresponding author, K. H., upon reasonable request.

References

- Angelopoulos, S., E. Bendoly, J. Fransoo, K. Hoberg, C. Ou, and A. Tenhiälä. 2023. "Digital Transformation in Operations Management: Fundamental Change Through Agency Reversal." *Journal of Operations Management* 69 (6): 876–889. <https://doi.org/10.1002/joom.1271>.
- Bannert, M., P. Reimann, and C. Sonnenberg. 2014. "Process Mining Techniques for Analysing Patterns and Strategies in Students' Self-Regulated Learning." *Metacognition and Learning* 9 (2): 161–185. <https://doi.org/10.1007/s11409-013-9107-6>.
- Chatfield, D. C. 2013. "Underestimating the Bullwhip Effect: A Simulation Study of the Decomposability Assumption." *International Journal of Production Research* 51 (1): 230–244. <https://doi.org/10.1080/00207543.2012.660576>.
- Choueiri, A. C., D. M. V. Sato, E. E. Scalabrin, and E. A. P. Santos. 2020. "An Extended Model for Remaining Time Prediction in Manufacturing Systems Using Process Mining." *Journal of Manufacturing Systems* 56:188–201. <https://doi.org/10.1016/j.jmsy.2020.06.003>.
- Choy, K.-W., R. C. Gazzo, and R. Panikkar. 2024. *Better Together: Process and Task Mining, a Powerful AI Combo*. McKinsey. <https://www.mckinsey.com/capabilities/operations/our-insights/better-together-process-and-task-mining-a-powerful-ai-combo#/>
- Cooper, M. C., D. M. Lambert, and J. D. Pagh. 1997. "Supply Chain Management: More Than a New Name for Logistics." *The International Journal of Logistics Management* 8 (1): 1–14. <https://doi.org/10.1108/09574099710805556>.
- Denisov, V., D. Fahland, and W. M. P. Van Der Aalst. 2018. "Unbiased, Fine-Grained Description of Processes Performance from Event Data." In *Business Process Management* (Vol. 11080), edited by M. Weske, M. Montali, I. Weber, and J. Vom Brocke, 139–157. Cham: Springer. https://doi.org/10.1007/978-3-319-98648-7_9.
- Di Francescomarino, C., C. Ghidini, F. M. Maggi, and F. Milani. 2018. *Predictive Process Monitoring Methods: Which One Suits Me Best?* (arXiv:1804.02422). arXiv. <http://arxiv.org/abs/1804.02422>.
- Dolgui, A., D. Ivanov, and B. Sokolov. 2018. "Ripple Effect in the Supply Chain: An Analysis and Recent Literature." *International Journal of Production Research* 56 (1–2): 414–430. <https://doi.org/10.1080/00207543.2017.1387680>.
- Fildes, R., P. Goodwin, M. Lawrence, and K. Nikolopoulos. 2009. "Effective Forecasting and Judgmental Adjustments: An Empirical Evaluation and Strategies for Improvement in Supply-Chain Planning." *International Journal of Forecasting* 25 (1): 3–23. <https://doi.org/10.1016/j.ijforecast.2008.11.010>.
- Friederich, J., D. P. Francis, S. Lazarova-Molnar, and N. Mohamed. 2022. "A Framework for Data-Driven Digital Twins of Smart Manufacturing Systems." *Computers in Industry* 136:103586. <https://doi.org/10.1016/j.compind.2021.103586>.
- Fu, G., J. Wang, and M. Yan. 2016. "Anatomy of Tianjin Port Fire and Explosion: Process and Causes." *Process Safety Progress* 35 (3): 216–220. <https://doi.org/10.1002/prs.11837>.
- Garcia, C. D. S., A. Meinheim, E. R. Faria Junior, M. R. Dallagassa, D. M. V. Sato, D. R. Carvalho, E. A. P. Santos, and E. E. Scalabrin. 2019. "Process Mining Techniques and Applications – A Systematic Mapping Study." *Expert Systems with Applications* 133:260–295. <https://doi.org/10.1016/j.eswa.2019.05.003>.
- Guha, S., and S. Kumar. 2018. "Emergence of Big Data Research in Operations Management, Information Systems,

- and Healthcare: Past Contributions and Future Roadmap.” *Production and Operations Management* 27 (9): 1724–1735. <https://doi.org/10.1111/poms.12833>.
- Gupta, S., S. Modgil, and A. Gunasekaran. 2020. “Big Data in Lean six Sigma: A Review and Further Research Directions.” *International Journal of Production Research* 58 (3): 947–969. <https://doi.org/10.1080/00207543.2019.1598599>.
- Ho, G. T. S., H. C. W. Lau, S. K. Kwok, C. K. M. Lee, and W. Ho. 2009. “Development of a Co-Operative Distributed Process Mining System for Quality Assurance.” *International Journal of Production Research* 47 (4): 883–918. <https://doi.org/10.1080/00207540701441939>.
- Hofmann, E. 2017. “Big Data and Supply Chain Decisions: The Impact of Volume, Variety and Velocity Properties on the Bullwhip Effect.” *International Journal of Production Research* 55 (17): 5108–5126. <https://doi.org/10.1080/00207543.2015.1061222>.
- Hosseini, S., D. Ivanov, and A. Dolgui. 2019. “Review of Quantitative Methods for Supply Chain Resilience Analysis.” *Transportation Research Part E: Logistics and Transportation Review* 125:285–307. <https://doi.org/10.1016/j.tre.2019.03.001>.
- Imdahl, C., W. Schmidt, and K. Hoberg. 2022. “Sustained Learning Under Algorithm-Driven Automation.” SSRN Electronic Journal. <https://doi.org/10.2139/ssrn.4292438>
- Ivanov, D. 2017. “Simulation-based Ripple Effect Modelling in the Supply Chain.” *International Journal of Production Research* 55 (7): 2083–2101. <https://doi.org/10.1080/00207543.2016.1275873>.
- Ivanov, D. 2024. “Conceptualisation of a 7-Element Digital Twin Framework in Supply Chain and Operations Management.” *International Journal of Production Research* 62 (6): 2220–2232. <https://doi.org/10.1080/00207543.2023.2217291>.
- Ivanov, D., A. Dolgui, and B. Sokolov. 2019. “The Impact of Digital Technology and Industry 4.0 on the Ripple Effect and Supply Chain Risk Analytics.” *International Journal of Production Research* 57 (3): 829–846. <https://doi.org/10.1080/00207543.2018.1488086>.
- Jans, M., J. M. Van Der Werf, N. Lybaert, and K. Vanhoof. 2011. “A Business Process Mining Application for Internal Transaction Fraud Mitigation.” *Expert Systems with Applications* 38 (10): 13351–13359. <https://doi.org/10.1016/j.eswa.2011.04.159>.
- Karlsson, C., ed. 2016. *Research Methods for Operations Management*. 2nd ed. New York: Routledge.
- Katsaliaki, K., P. Galetsi, and S. Kumar. 2022. “Supply Chain Disruptions and Resilience: A Major Review and Future Research Agenda.” *Annals of Operations Research* 319 (1): 965–1002. <https://doi.org/10.1007/s10479-020-03912-1>.
- Khosrowabadi, N., K. Hoberg, and C. Imdahl. 2022. “Evaluating Human Behaviour in Response to AI Recommendations for Judgemental Forecasting.” *European Journal of Operational Research* 303 (3): 1151–1167. <https://doi.org/10.1016/j.ejor.2022.03.017>.
- Kim, Y., Y. Chen, and K. Linderman. 2015. “Supply Network Disruption and Resilience: A Network Structural Perspective.” *Journal of Operations Management* 33–34 (1): 43–59. <https://doi.org/10.1016/j.jom.2014.10.006>.
- Knoll, D., G. Reinhart, and M. Pröglmeier. 2019. “Enabling Value Stream Mapping for Internal Logistics Using Multi-dimensional Process Mining.” *Expert Systems with Applications* 124:130–142. <https://doi.org/10.1016/j.eswa.2019.01.026>.
- Lee, C. K. H., G. T. S. Ho, K. L. Choy, and G. K. H. Pang. 2014. “A RFID-Based Recursive Process Mining System for Quality Assurance in the Garment Industry.” *International Journal of Production Research* 52 (14): 4216–4238. <https://doi.org/10.1080/00207543.2013.869632>.
- Lorenz, R., J. Senoner, W. Sihn, and T. Netland. 2021. “Using Process Mining to Improve Productivity in Make-to-Stock Manufacturing.” *International Journal of Production Research* 59 (16): 4869–4880. <https://doi.org/10.1080/00207543.2021.1906460>.
- Lugaresi, G., A. D. Ciappina, M. Rossi, and A. Matta. 2023. “Exploiting a Combined Process Mining Approach to Enhance the Discovery and Analysis of Support Processes in Manufacturing.” *International Journal of Computer Integrated Manufacturing* 36 (2): 169–189. <https://doi.org/10.1080/0951192X.2022.2090024>.
- Manavalan, E., and K. Jayakrishna. 2019. “A Review of Internet of Things (IoT) Embedded Sustainable Supply Chain for Industry 4.0 Requirements.” *Computers & Industrial Engineering* 127:925–953. <https://doi.org/10.1016/j.cie.2018.11.030>.
- Munoz, A., and M. Dunbar. 2015. “On the Quantification of Operational Supply Chain Resilience.” *International Journal of Production Research* 53 (22): 6736–6751. <https://doi.org/10.1080/00207543.2015.1057296>.
- Partington, A., M. Wynn, S. Suriadi, C. Ouyang, and J. Karnon. 2015. “Process Mining for Clinical Processes: A Comparative Analysis of Four Australian Hospitals.” *ACM Transactions on Management Information Systems* 5 (4): 1–18. <https://doi.org/10.1145/2629446>.
- Perera, H. N., J. Hurley, B. Fahimnia, and M. Reisi. 2019. “The Human Factor in Supply Chain Forecasting: A Systematic Review.” *European Journal of Operational Research* 274 (2): 574–600. <https://doi.org/10.1016/j.ejor.2018.10.028>.
- Polato, M., A. Sperduti, A. Burattin, and M. D. Leoni. 2018. “Time and Activity Sequence Prediction of Business Process Instances.” *Computing* 100 (9): 1005–1031. <https://doi.org/10.1007/s00607-018-0593-x>.
- Pourbafrani, M., S. J. Van Zelst, and W. M. P. Van Der Aalst. 2020. “Supporting Automatic System Dynamics Model Generation for Simulation in the Context of Process Mining.” In *Business Information Systems* (Vol. 389), edited by W. Abramowicz, and G. Klein, 249–263. Cham: Springer. https://doi.org/10.1007/978-3-030-53337-3_19.
- Rother, M., and J. Shook. 2018. *Learning to see: Value-Stream Mapping to Create Value and Eliminate Muda (Version 1.5)*. 20th Anniversary Edition. Cambridge: Lean Enterprise Inst.
- Sánchez, M., E. Exposito, and J. Aguilar. 2020. “Implementing Self-* Autonomic Properties in Self-Coordinated Manufacturing Processes for the Industry 4.0 Context.” *Computers in Industry* 121:103247. <https://doi.org/10.1016/j.compind.2020.103247>.
- Sodhi, M. S., and C. S. Tang. 2019. “Research Opportunities in Supply Chain Transparency.” *Production and Operations Management* 28 (12): 2946–2959. <https://doi.org/10.1111/poms.13115>.
- Sunk, A., P. Kuhlang, T. Edtmayr, and W. Sihn. 2017. “Developments of Traditional Value Stream Mapping to Enhance Personal and Organisational System and Methods Competencies.” *International Journal of Production Research*

- 55 (13): 3732–3746. <https://doi.org/10.1080/00207543.2016.1272764>.
- Tienemaa, I., M. Dumas, M. La Rosa, and F. M. Maggi. 2018. *Outcome-Oriented Predictive Process Monitoring: Review and Benchmark* (arXiv:1707.06766). arXiv. <http://arxiv.org/abs/1707.06766>.
- Tiwari, M. K., B. Bidanda, J. Geunes, K. Fernandes, and A. Dolgui. 2024. “Supply Chain Digitisation and Management.” *International Journal of Production Research* 62 (8): 2918–2926. <https://doi.org/10.1080/00207543.2024.2316476>.
- Tozanli, Ö., and M. J. Saénz. 2022. “Unlocking the Potential of Digital Twins in Supply Chains.” *MIT Sloan Management Review* 63 (4): 1–4.
- Van Der Aalst, W. M. P. 2016. *Process Mining: Data Science in Action*. Berlin Heidelberg: Springer. <https://doi.org/10.1007/978-3-662-49851-4>.
- Van Der Aalst, W. M. P. 2018. *Proceedings of the 50th Computer Simulation Conference*. San Diego: Society for Computer Simulation International.
- Van Der Aalst, W. M. P. 2019. “Object-Centric Process Mining: Dealing with Divergence and Convergence in Event Data.” In *Software Engineering and Formal Methods* (Vol. 11724), edited by P. C. Ölveczky, and G. Salaün, 3–25. Cham: Springer. https://doi.org/10.1007/978-3-030-30446-1_1.
- Van Der Aalst, W. M. P. 2021. “Federated Process Mining: Exploiting Event Data Across Organizational Boundaries.” In *2021 IEEE International Conference on Smart Data Services (SMDS)*, Chicago, IL, USA.
- Van Der Aalst, W. M. P., A. Adriansyah, A. K. A. De Medeiros, F. Arcieri, T. Baier, T. Blickle, J. C. Bose, ... M. Wynn. 2012. “Process Mining Manifesto.” In *Business Process Management Workshops* (Vol. 99), edited by F. Daniel, K. Barkaoui, and S. Dustdar, 169–194. Berlin Heidelberg: Springer. https://doi.org/10.1007/978-3-642-28108-2_19.
- Van Der Aalst, W. M. P., V. Rubin, H. M. W. Verbeek, B. F. Van Dongen, E. Kindler, and C. W. Günther. 2010. “Process Mining: A Two-Step Approach to Balance Between Underfitting and Overfitting.” *Software & Systems Modeling* 9 (1): 87–111. <https://doi.org/10.1007/s10270-008-0106-z>.
- Van Der Aalst, W. M. P., M. H. Schonenberg, and M. Song. 2011. “Time Prediction Based on Process Mining.” *Information Systems* 36 (2): 450–475. <https://doi.org/10.1016/j.is.2010.09.001>.
- Van Zelst, S. J., F. Mannhardt, M. De Leoni, and A. Koschmider. 2021. “Event Abstraction in Process Mining: Literature Review and Taxonomy.” *Granular Computing* 6 (3): 719–736. <https://doi.org/10.1007/s41066-020-00226-2>.
- Wang, R., and O. R. Zaiane. 2015. “Discovering Process in Curriculum Data to Provide Recommendation.” In *Proceedings of the 5th International Conference on Educational Data Mining*, 580–581. Massachusetts: International Educational Data Mining Society (IEDMS).
- Wangphanich, P., S. Kara, and B. Kayis. 2010. “Analysis of the Bullwhip Effect in Multi-Product, Multi-Stage Supply Chain Systems—a Simulation Approach.” *International Journal of Production Research* 48 (15): 4501–4517. <https://doi.org/10.1080/00207540902950852>.
- Werner, M., M. Wiese, and A. Maas. 2021. “Embedding Process Mining Into Financial Statement Audits.” *International Journal of Accounting Information Systems* 41:100514. <https://doi.org/10.1016/j.accinf.2021.100514>.
- Yoo, S., M. Cho, E. Kim, S. Kim, Y. Sim, D. Yoo, H. Hwang, and M. Song. 2016. “Assessment of Hospital Processes Using a Process Mining Technique: Outpatient Process Analysis at a Tertiary Hospital.” *International Journal of Medical Informatics* 88:34–43. <https://doi.org/10.1016/j.ijmedinf.2015.12.018>.

Appendix 1. Expert interviews

#	Round	Duration	Industry	Current role	Years of experience
1	1	00:43h	Strategy consulting	Expert Associate Partner	10
2	1	01:00h	Semiconductor	Operations Intelligence Manager	8
3	1	00:54h	Chemical	Process Mining Analyst	2
4	1	00:59h	Chemical	Process Manager	8
5	1	00:45h	Pharma & chemical	Head Global Supply Chain Process & Performance Management	20
6	1	00:54h	Semiconductor	Director IT Innovation & Data Driven Insights	11
7	1	01:00h	Process mining	Account Executive Manufacturing	5
8	1	00:38h	Automotive	Process Manager	5
9	1	00:32h	Care products	Manager Innovation Process Automation	5
10	1	01:02h	Semiconductor	Senior Supply Chain Engineer	6
11	1	00:56h	Chemical	Head of Business Process & Data Quality Management	20+
12	1	01:17h	Process mining	Vice President Products Solution Engineering	12
13	1	00:36h	Steel & metal	CEO	16
14	1	00:58h	Process mining	General Manager Supply Chain	6
15	1	00:48h	Semiconductor	Supply Chain Analytics Product Owner	7
16	1	00:40h	Process mining	Co-founder	15
17	1	00:44h	Process mining	Head of Operations	5
18	1	00:59h	Appliances	Global Planning & Logistics Manager	9
19	1	00:56h	Rubber & sealing	Director Global Supply Chain Controlling	17
20	1	00:56h	Process mining	Product Manager Supply Chain	7
21	1	00:40h	Education & research	Professor of Operations and Logistics Management	25+
22	2	01:38h	Chemical	Head of Business Process & Data Quality Management	20+
23	2	01:57h	Process mining	General Manager Supply Chain	6
24	2	01:00h	Semiconductor	Director IT Innovation & Data Driven Insights	11
25	2	01:00h	Process mining	Vice President Products Solution Engineering	12
26	2	01:30h	Rubber & sealing	Director Global Supply Chain Controlling	17